Human-Like Redundancy Resolution for Real-Time Bimanual Humanoid Teleoperation

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***Abstract*— Real-time bimanual humanoid teleoperation requires accurate wrist tracking and stable arm posture. Kinematic redundancy allows different elbow configurations for the same wrist pose, so wrist-only tracking may cause elbow drift or configuration jumps. Anthropomorphic constraints help guide elbow posture, but fixed constraints can restrict redundant motion and increase the risk of kinematic singularities. We propose a human-like redundancy-resolution framework that combines adaptive elbow guidance with singularity-robust wrist tracking. Shoulder–wrist geometric retargeting generates robot-scale wrist targets. An adaptive elbow-offset criterion uses offset direction to guide posture and its magnitude to regulate guidance strength, progressively releasing redundancy as the human arm straightens. A hierarchical velocity solver prioritizes singularity-robust wrist tracking and applies adaptive elbow guidance in the wrist-task null space. In a fixed-wrist-target ablation, adaptive guidance reduces mean elbow-direction error by 54.4% compared with removing this guidance. In a workspace-boundary stress test, singularity-robust tracking reduces peak raw joint-velocity demand by 97.9% compared with the undamped pseudoinverse. Real-world bimanual tasks and cross-platform retargeting demonstrate applicability across robot morphologies.**

I. Introduction

Real-time bimanual humanoid teleoperation enables operators to transfer human motion to robots for dexterous manipulation and demonstration collection [2]–[4]. Beyond reproducing wrist trajectories, effective teleoperation requires robot arms to maintain natural configurations and respond smoothly across diverse motions. For a 7-degree-of-freedom arm, however, a six-dimensional wrist task generally leaves one redundant degree of freedom. Resolving this redundancy is essential to achieving predictable arm behavior while preserving the operator's end-effector intent.

Existing approaches address this problem through wrist tracking or additional posture guidance. Wrist-only inverse kinematics leaves elbow posture underconstrained, potentially causing elbow drift or abrupt configuration changes between nearby wrist targets (the top left part of Fig. 1). Anthropomorphic retargeting incorporates human keypoints, geometric redundancy parameters, or posture objectives to improve human–robot configuration consistency [5], [7], [11], [13], [15]. However, rigidly enforcing an anthropomorphic target can restrict redundant configuration adjustments near singularities, compromising wrist tracking or rendering the combined objectives infeasible (the bottom left part of Fig. 1). Moreover, the reliability of human posture information varies with arm geometry, making fixed-strength guidance inappropriate across different configurations.

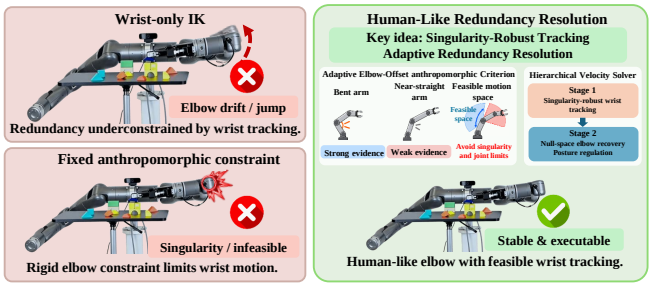


Fig. 1. Human-like redundancy resolution. Wrist-only inverse kinematics leaves the elbow underconstrained, whereas a fixed anthropomorphic objective can become unreliable near full extension. Our method adapts elbow evidence and combines it with singularity-robust wrist tracking.

This reliability issue becomes particularly pronounced near full arm extension. As the human shoulder, elbow, and wrist approach collinearity, the elbow's lateral offset from the shoulder–wrist axis vanishes, and its normalized direction becomes sensitive to measurement noise. Strongly enforcing this direction can therefore induce unnecessary posture corrections when the observation is least informative. Meanwhile, poor conditioning of the robot wrist Jacobian can amplify modest task-space commands into large joint-velocity requests. These are distinct challenges: reducing unreliable elbow guidance does not itself resolve robot singularities, while regularizing wrist execution does not determine a human-like elbow posture.

To address both challenges, we propose a human-like redundancy-resolution framework that prioritizes singularity-robust wrist tracking while adapting elbow guidance to the reliability of human posture observations (the right part of Fig. 1). Shoulder–wrist geometric retargeting generates robot-scale wrist targets, while an adaptive elbow-offset anthropomorphic criterion uses the offset direction to guide posture and its magnitude to progressively weaken and deactivate guidance near full arm extension. Finally, a hierarchical velocity solver suppresses near-singular joint-velocity amplification and applies adaptive posture guidance in the wrist-task null space. Experiments show that the framework improves elbow-posture consistency and suppresses excessive near-singular velocity demands while maintaining accurate wrist tracking. Cross-platform retargeting and real-world bimanual tasks further demonstrate its practical applicability.

The main contributions are:

[1]Code: www.github.com/chaotan2003/Bimanual-Teleoperation
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- **Adaptive anthropomorphic redundancy resolution.** We introduce an elbow-offset criterion that continuously adjusts anthropomorphic guidance according to the reliability of human posture geometry. This enables a smooth transition from posture guidance in bent-arm configurations to redundancy release near full extension.
- **Singularity-robust task-priority coordination.** We integrate adaptive elbow guidance with J-PARSE-based wrist tracking in a task-priority framework. It suppresses near-singular joint-velocity amplification and uses redundancy to improve arm posture while giving the wrist task priority at the velocity level.
- **Experimental and cross-platform validation.** Adding AEAC reduces mean elbow-direction error by 54.4% compared with removing it in fixed-wrist ablations, and singularity-robust tracking reduces peak raw joint-velocity demand by 97.9% relative to the undamped pseudoinverse in boundary stress tests. Qualitative cross-platform retargeting and real-robot bimanual tasks demonstrate practical feasibility.

II. Related Work

A. Bimanual Humanoid Teleoperation

Recent bimanual teleoperation systems transfer operator hand motion to humanoid platforms via immersive visual feedback or wearable interfaces [3], [4], [41], with whole-body extensions coordinating torso and legs in real time [1], [27], [28], [29] and cross-platform frameworks integrating sensing, retargeting, and execution [18], [26], [31]. End-effector retargeting preserves task-space intent but leaves the elbow in the robot null space [14], [19], while keypoint-based methods add shoulder and elbow landmarks for richer posture information [8], [15] yet degrade near full extension or self-occlusion. Our method uses the same human keypoints for redundancy guidance, but adapts their influence according to the geometric observability of the elbow and preserves wrist tracking through task-priority control.

B. Anthropomorphic Retargeting

Geometric redundancy parameterizations [5], [9], closed-form solvers [7], and posture-constrained mappings [10], [11] address the elbow redundancy that remains after the wrist pose is fixed. Other approaches encode anthropomorphic behavior through monocular arm imitation [12], [13], hybrid redundancy resolution [21], or object-aware posture selection [22], [23]. Arm-plane-normal matching [11] represents anthropomorphic posture through a unit direction. Our AEAC couples directional alignment with the elbow's perpendicular offset magnitude, making guidance strength dependent on human arm geometry. This extends direction-only matching into a geometry-adaptive redundancy objective that regulates both posture alignment and its activation.

C. Analytical Geometric Redundancy Solvers

Manipulability measures [32], [34], [42] and redundancy-resolution frameworks [33], [36] quantify singularity proximity and exploit the null space for secondary objectives; for 7-DoF arms, the wrist task leaves one degree of freedom that can be controlled through null-space motion [5], [7]. Singularity-robust damping and task-priority mappings [9], [14], [35] improve numerical behavior near poorly conditioned configurations, while hierarchical formulations [38], [39] explicitly preserve primary task priorities. Our method follows this task-priority principle with a J-PARSE velocity mapping for the stacked wrist task and uses adaptive elbow guidance only in the resulting null space.

III. Methodology

We develop a real-time motion-retargeting and hierarchical velocity-solving framework for bimanual teleoperation with two 7-DoF humanoid arms. Given XR shoulder, elbow, and wrist keypoints together with controller poses, the system generates robot wrist targets through geometric retargeting, resolves arm redundancy with an adaptive elbow-offset anthropomorphic criterion, and computes joint commands using a hierarchical velocity solver with singularity-robust wrist tracking. As shown in Fig. 2, the method comprises four components: (i) Shoulder–Wrist Geometric Retargeting (SWGR, Section III-B), (ii) the adaptive elbow-offset anthropomorphic criterion (AEAC, Section III-C), (iii) singularity-robust wrist tracking (Section III-D), and (iv) the hierarchical velocity solver (Section III-E).

A. Problem Formulation

We consider a humanoid robot with two 7-DoF arms, whose joint configuration is denoted by $\mathbf{q} = [\mathbf{q}_L^\mathsf{T}\ \mathbf{q}_R^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^{14}$, with $\mathbf{q}_i \in \mathbb{R}^7$. At time t, the XR interface provides

$$\mathcal{X}_t = \left\{\mathbf{p}^h_{s,i}, \mathbf{p}^h_{e,i}, \mathbf{p}^h_{w,i}, \mathbf{R}^h_{c,i}\right\}_{i\in\{L,R\}}, \tag{1}$$

where $\mathbf{p}^h_{s,i}$, $\mathbf{p}^h_{e,i}$, and $\mathbf{p}^h_{w,i}$ are human shoulder, elbow, and wrist positions, and $\mathbf{R}^h_{c,i}$ is the controller orientation. Given the retargeted wrist velocity $\mathbf{v}_d$, we compute a singularity-robust primary wrist velocity

$$\dot{\mathbf{q}}_{\text{task}} = \mathbf{J}^{\#}_{\text{JPARSE}}\mathbf{v}_d, \tag{2}$$

and project secondary posture commands into the null space of the safety Jacobian:

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}_{\text{task}} + \mathbf{N}\dot{\mathbf{q}}_{\text{sec}}, \tag{3}$$

where $\mathbf{N}$ is the J-PARSE task null-space projector and $\dot{\mathbf{q}}_{\text{sec}}$ contains AEAC and posture-regulation terms. With solver period Δt, the state update is

$$\mathbf{q}_{t+1} = \text{clip}(\mathbf{q}_t + \dot{\mathbf{q}}_t\Delta t,\ \mathbf{q}_{\min}, \mathbf{q}_{\max}). \tag{4}$$

B. Shoulder–Wrist Geometric Retargeting

We use Shoulder–Wrist Geometric Retargeting (SWGR, Fig. 2(I)) to convert human upper-arm motion into robot-scale wrist targets. Let $\mathbf{p}^h_s$ and $\mathbf{p}^h_w$ denote the human shoulder and wrist positions, $\mathbf{p}^r_s$ the corresponding robot shoulder anchor, and $\mathbf{R}_{rh}$ the rotation from the human tracking frame to the robot frame. The desired wrist position is

$$\mathbf{p}^r_{w,d} = \mathbf{p}^r_s + \alpha\mathbf{R}_{rh}\left(\mathbf{p}^h_w - \mathbf{p}^h_s\right), \tag{5}$$

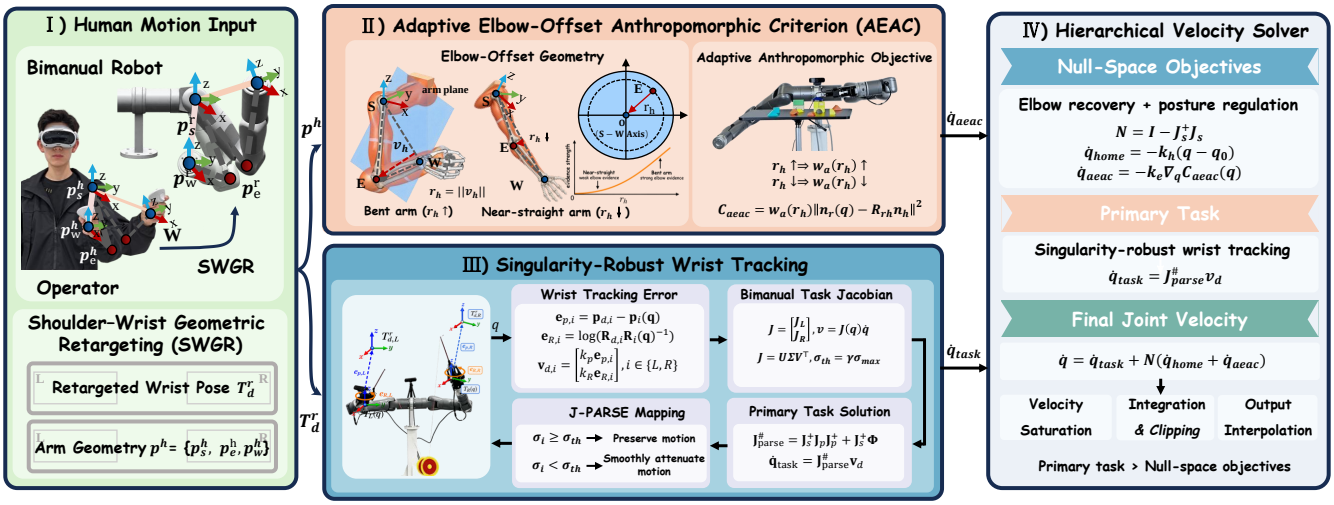


Fig. 2. **Framework of Human-Like Redundancy Resolution for Real-Time Bimanual Humanoid Teleoperation.** I) *Shoulder–Wrist Geometric Retargeting (SWGR)*: XR shoulder, elbow, and wrist keypoints with controller poses are retargeted to robot-scale wrist targets. II) *Adaptive Elbow-Offset Anthropomorphic Criterion (AEAC)*: the elbow-offset direction guides redundancy with adaptive weighting. III) *Singularity-Robust Wrist Tracking*: J-PARSE regularizes the stacked wrist Jacobian. IV) *Hierarchical Velocity Solver*: the primary wrist task is combined with AEAC and posture regulation in the task null space to generate joint commands.

where $\alpha = l_r/l_h$, and l_h and l_r are the human and robot reference shoulder–wrist lengths. Because only a relative shoulder–wrist vector is used, translations of the XR world origin do not affect the retargeted position.

The controller orientation is mapped using a calibrated tool offset according to $\mathbf{R}_d^r = \mathbf{R}_{rh}\mathbf{R}_c^h\mathbf{R}_{\text{off}}$. The desired wrist pose is therefore

$$\mathbf{T}_d = \begin{bmatrix} \mathbf{R}_d^r & \mathbf{p}_{w,d}^r \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}. \tag{6}$$

Before velocity solving, each target position is smoothed by a One Euro filter [46]. The target orientation is filtered by quaternion spherical interpolation [47], which avoids directly averaging rotation matrices. These filters act on the desired pose only; they do not alter the task hierarchy or the null-space computation.

C. Adaptive Elbow-Offset Anthropomorphic Criterion

The 7-DoF wrist task leaves one redundant degree of freedom per arm. The Adaptive Elbow-Offset Anthropomorphic Criterion (AEAC, Fig. 2(II)) uses the human elbow observation to guide this redundancy while reducing its influence when the elbow direction is poorly observable. Fixed-weight constraints [9], [11] can amplify tracking noise near full extension, where the shoulder, elbow, and wrist become nearly collinear. AEAC separates the elbow evidence into a direction and a magnitude: the direction defines the anthropomorphic objective, and the magnitude determines how strongly that objective is used.

Anthropomorphic criterion. For one arm, let $\mathbf{p}_s^h$, $\mathbf{p}_e^h$, and $\mathbf{p}_w^h$ denote the human shoulder, elbow, and wrist positions in a common frame. The shoulder–wrist axis direction is

$$\mathbf{d}_h = \frac{\mathbf{p}_w^h - \mathbf{p}_s^h}{\max(\|\mathbf{p}_w^h - \mathbf{p}_s^h\|_2, \varepsilon)}. \tag{7}$$

The projection of the elbow onto this axis and the perpendicular offset are

$$\mathbf{p}_{\text{proj}}^h = \mathbf{p}_s^h + \mathbf{d}_h\mathbf{d}_h^\mathsf{T}\left(\mathbf{p}_e^h - \mathbf{p}_s^h\right). \tag{8}$$

The perpendicular elbow offset is

$$\mathbf{v}_h = \mathbf{p}_e^h - \mathbf{p}_{\text{proj}}^h. \tag{9}$$

The human elbow direction is

$$\mathbf{n}_h = \frac{\mathbf{v}_h}{\max(\|\mathbf{v}_h\|_2, \varepsilon)}, \tag{10}$$

and its offset magnitude is denoted in the following by $h = \|\mathbf{v}_h\|_2$. The robot direction $\mathbf{n}_r(\mathbf{q})$ is computed by applying the same construction to the forward-kinematic positions $\mathbf{p}_s^r(\mathbf{q})$, $\mathbf{p}_e^r(\mathbf{q})$, and $\mathbf{p}_w^r(\mathbf{q})$. After transforming the human direction into the robot frame, the directional anthropomorphic loss is

$$\mathcal{L}_a(\mathbf{q}) = \|\mathbf{n}_r(\mathbf{q}) - \mathbf{R}_{rh}\mathbf{n}_h\|_2^2. \tag{11}$$

Adaptive implementation. The length h measures how far the elbow lies from the shoulder–wrist axis and therefore how reliable the elbow direction is. As the arm straightens, $h \to 0$ and $\mathbf{n}_h$ becomes noise-sensitive. With a deadband δ and maximum weight $w_{\max}$, the adaptive weight is

$$\omega_a = w_{\max}\,\text{clip}\left(\frac{h-\delta}{l_h - \delta}, 0, 1\right). \tag{12}$$

The final AEAC cost is

$$C_{\text{AEAC}}(\mathbf{q}) = \omega_a\mathcal{L}_a(\mathbf{q}). \tag{13}$$

When $h > \varepsilon$, differentiating the normalized direction gives

$$\frac{\partial \mathbf{n}_h}{\partial \mathbf{v}_h} = \frac{\mathbf{I} - \mathbf{n}_h\mathbf{n}_h^\mathsf{T}}{h}, \tag{14}$$

so the sensitivity of the normalized elbow direction scales inversely with the offset magnitude. AEAC therefore uses this magnitude to regulate the influence of directional alignment, retaining posture guidance when the geometry is informative and deactivating it within the extension deadband. When $h \leq \delta$, $\omega_a = 0$ and the elbow observation is ignored; when the arm has a measurable lateral offset, the criterion is gradually activated. In the implementation, the gradient of Eq. (13) is evaluated by automatic differentiation and used as a null-space velocity. AEAC is therefore a soft secondary preference, rather than a hard constraint that can override wrist tracking.

D. Singularity-Robust Wrist Tracking

Motivated by J-PARSE [43], which adapts task-space commands to the available mobility near singularities, we combine singularity-robust wrist tracking (Fig. 2(III)) with AEAC-based anthropomorphic redundancy resolution. For each wrist, the geometric Jacobian maps joint velocity to a six-dimensional twist; the two Jacobians are stacked as

$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} \mathbf{J}_L(\mathbf{q}) \\ \mathbf{J}_R(\mathbf{q}) \end{bmatrix}, \qquad \mathbf{v} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}. \tag{15}$$

For arm i, define the position and orientation errors as

$$\mathbf{e}_{p,i} = \mathbf{p}_{d,i} - \mathbf{p}_i(\mathbf{q}), \qquad \mathbf{e}_{R,i} = \mathrm{Log}\big(\mathbf{R}_{d,i}\mathbf{R}_i(\mathbf{q})^{-1}\big). \tag{16}$$

The desired stacked twist is

$$\mathbf{v}_d = \begin{bmatrix} k_p\mathbf{e}_{p,L}^\mathsf{T} & k_R\mathbf{e}_{R,L}^\mathsf{T} & k_p\mathbf{e}_{p,R}^\mathsf{T} & k_R\mathbf{e}_{R,R}^\mathsf{T} \end{bmatrix}^\mathsf{T}. \tag{17}$$

The J-PARSE construction is defined as follows. We compute the singular value decomposition

$$\mathbf{J} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\mathsf{T}. \tag{18}$$

Let $\sigma_{\max}$ denote the largest singular value and set

$$\sigma_{\mathrm{th}} = \gamma\sigma_{\max}. \tag{19}$$

Following J-PARSE [43], let $\mathbf{u}_j$ and $\mathbf{v}_j$ denote the corresponding left and right singular vectors of the stacked wrist Jacobian. The safety Jacobian is

$$\mathbf{J}_s = \sum_{j=1}^{12} \max(\sigma_j, \sigma_{\mathrm{th}})\mathbf{u}_j\mathbf{v}_j^\mathsf{T}. \tag{20}$$

The inverse gain for each singular direction is

$$g_j = \begin{cases} 1/\sigma_j, & \sigma_j > \sigma_{\mathrm{th}}, \\ \sigma_j/\sigma_{\mathrm{th}}^2, & \sigma_j \leq \sigma_{\mathrm{th}}. \end{cases} \tag{21}$$

The singularity-robust inverse is therefore

$$\mathbf{J}^{\#}_{\mathrm{JPARSE}} = \sum_{j=1}^{12} g_j\mathbf{v}_j\mathbf{u}_j^\mathsf{T}. \tag{22}$$

This mapping preserves the inverse gains of well-conditioned directions and attenuates poorly conditioned ones. The primary wrist-task velocity is then

$$\dot{\mathbf{q}}_{\mathrm{task}} = \mathbf{J}^{\#}_{\mathrm{JPARSE}}\mathbf{v}_d. \tag{23}$$

E. Hierarchical Velocity Solver

The primary wrist task is combined with secondary objectives only after the J-PARSE mapping has been computed (Fig. 2(IV)). The associated null-space projector is

$$\mathbf{N} = \mathbf{I} - \mathbf{J}_s^{+}\mathbf{J}_s. \tag{24}$$

The joint-centering velocity is

$$\dot{\mathbf{q}}_{\mathrm{home}} = -k_h(\mathbf{q} - \mathbf{q}_0), \tag{25}$$

where $\mathbf{q}_0$ is the reference configuration. The AEAC velocity is obtained from the negative gradient of the adaptive anthropomorphic cost:

$$\dot{\mathbf{q}}_{\mathrm{aeac}} = -k_e\nabla_{\mathbf{q}}C_{\mathrm{AEAC}}(\mathbf{q}). \tag{26}$$

The final command before limiting is

$$\dot{\mathbf{q}}^* = \dot{\mathbf{q}}_{\mathrm{task}} + \mathbf{N}\left(\dot{\mathbf{q}}_{\mathrm{home}} + \dot{\mathbf{q}}_{\mathrm{aeac}}\right). \tag{27}$$

The projected AEAC and posture terms reshape the arm configuration within the wrist-task null space, preserving the primary task at the differential level before command limiting. The combined velocity is then scaled to satisfy the joint-speed bound, integrated, and clipped to the joint limits.

The command is bounded by the configured maximum joint speed:

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}^* \min\left(1, \frac{\dot{q}_{\max}}{\|\dot{\mathbf{q}}^*\|_\infty + \varepsilon}\right). \tag{28}$$

It is then integrated and clipped to the robot joint limits using the configuration update in Section III-A. The default controller runs the velocity solver loop at 90 Hz with $\dot{q}_{\max} = 3.14$ rad/s. The position filter uses min_cutoff $= 5.0$ and $\beta = 10.0$, while the orientation filter uses $\alpha = 0.9$. Finally, the integrated joint targets are linearly interpolated at the 200 Hz output layer to improve command continuity without feeding interpolated values back into the solver state.

IV. Validation and Discussion

A. Experimental Setup and Evaluation Metrics

Compared methods. We compare three representative baselines: XR-Teleop [16] for wrist-centric IK, Fourier-Teleop [17] for posture-regularized differential IK, and GMR [15] for body-geometry-based retargeting. These baselines cover end-effector tracking, reference-posture regulation, and human-motion consistency. The comparison evaluates the benefits of geometry-adaptive elbow guidance for human-like redundancy resolution. We evaluate on 15 static key poses and three real-time motion sequences: elbow motion for anthropomorphic fidelity, circle motion for general motion range, and forward extension for extreme singular postures. Component ablations further isolate the contributions of AEAC and the singularity-robust mapping.

Experimental platform. Experiments run on a PICO 4 Ultra [44] or Vive Tracker [45] with a JAKA K1 humanoid dual-arm robot at 90 Hz (Ubuntu 24.04.4 LTS, Intel i7-11800H, NVIDIA RTX 3050 Laptop GPU). The

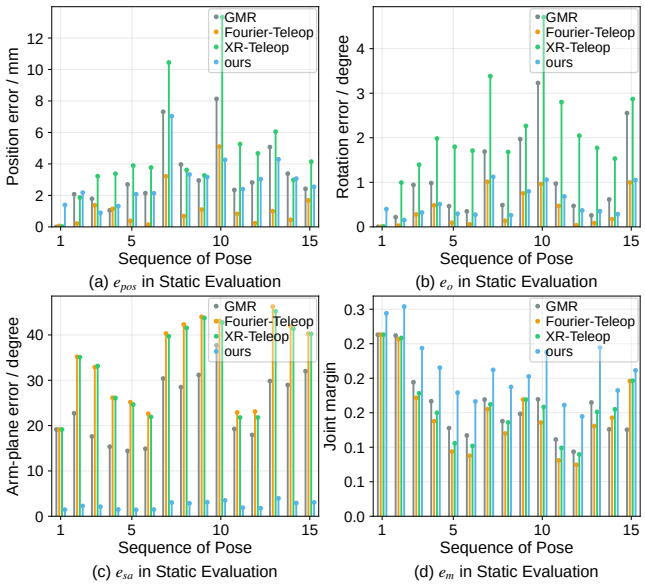


Fig. 3. Static evaluation of GMR, Fourier-Teleop, XR-Teleop, and Ours over 15 poses. (a) Wrist position error e_{pos}; (b) wrist rotation error e_o; (c) arm-plane error e_{sa}; and (d) joint margin e_m.

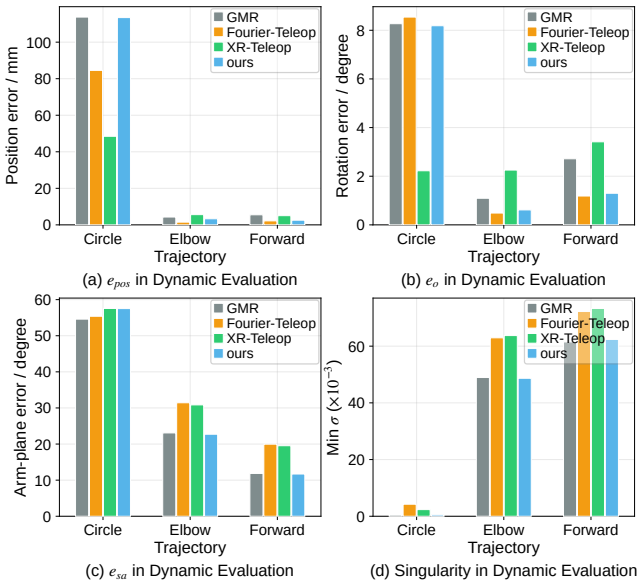


Fig. 4. Dynamic tracking comparison of GMR, Fourier-Teleop, XR-Teleop, and Ours on elbow, circle, and forward-extension trajectories. (a) Wrist position error e_{pos}; (b) wrist rotation error e_o; (c) arm-plane error e_{sa}; (d) singularity margin.

per-frame solve time averages 4.35 ms, within the 11.1 ms control budget.

Evaluation metrics. Functional tracking is measured by wrist position error $e_{\text{pos}} = \|\mathbf{p}^{\text{des}} - \mathbf{p}^{\text{cur}}\|_2$ and rotation error $e_o = \|\operatorname{Log}((\mathbf{R}^{\text{des}})^{\mathsf{T}}\mathbf{R}^{\text{cur}})\|_2$ [3], [15]. Anthropomorphic posture is measured by arm-plane error $e_{\text{sa}} = |\operatorname{atan2}(\sin(\psi^h - \psi^r), \cos(\psi^h - \psi^r))|$ [5], [7], [9], where ψ^h and ψ^r are signed swivel angles in the plane orthogonal to the shoulder–wrist axis; the time-series plots label the same metric as swivel-angle error. Joint safety is reported by the joint margin e_m [36], [37], and singularity behavior is evaluated separately [32], [35], [43].

B. Static Performance

Fig. 3 summarizes the static-pose results. The proposed method achieves the lowest arm-plane error across the 15 static poses, reaching $2.42 \pm 0.83°$, compared with $23.99 \pm 7.55°$ for GMR. This improvement in human–robot posture consistency is achieved with a wrist position error of 2.87 ± 1.52 mm and a rotation error of $0.53 \pm 0.33°$.

C. Dynamic Performance

Overall dynamic performance. Fig. 4 compares three trajectories: elbow motion, circle motion, and forward extension. The circle case is a limitation: XR-Teleop gives the lowest wrist errors (48.5 mm, 2.22°), while Ours gives 113.4 mm and 8.19° and does not improve arm-plane error (57.5°). On the elbow and forward-extension trajectories, however, Ours obtains the lowest arm-plane errors, 22.7° and 11.7°, with low wrist errors. The benefit is therefore targeted redundancy resolution and singularity-sensitive behavior, not uniformly best dynamic tracking.

Anthropomorphic behavior. Fig. 5(a) shows representative configurations from the elbow-motion sequence, and Fig. 5(b) plots the same arm-plane metric over time. Ours is best on elbow motion and forward extension. The circle trajectory is executed at high speed and drives the arms beyond the nominal workspace, serving primarily as a stability test rather than a posture-accuracy benchmark; the larger arm-plane error (57.52° versus 54.60° for GMR) reflects this trajectory-dependent trade-off. AEAC therefore helps when the elbow geometry is informative, but does not guarantee lower posture error for every motion pattern.

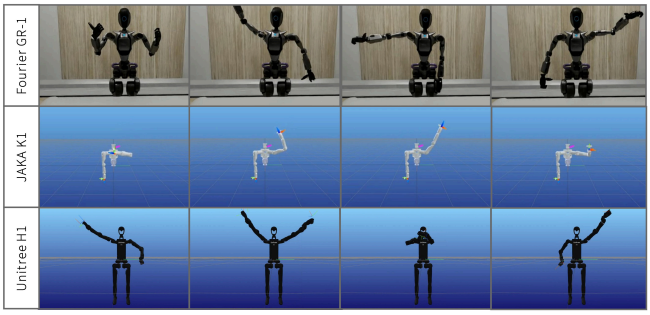
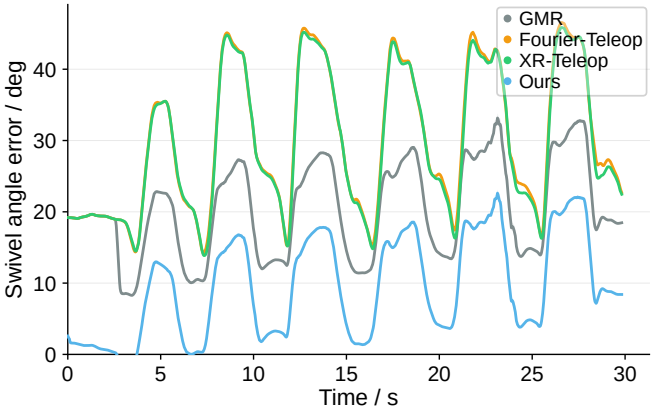


Fig. 6. Cross-platform retargeting on Fourier GR-1, JAKA K1, and Unitree H1 using identical parameters.

Singularity behavior. Fig. 7(b) shows that Ours maintains a higher and more stable singularity margin than the fixed-plane and human-release-only variants during repeated extension and recovery. Fig. 7(a) illustrates the associated forward-extension motion. The complete method remains executable near the workspace boundary without the abrupt joint motion observed for a rigid anthropomorphic constraint. Detailed component-level comparisons are given in Section IV-E.

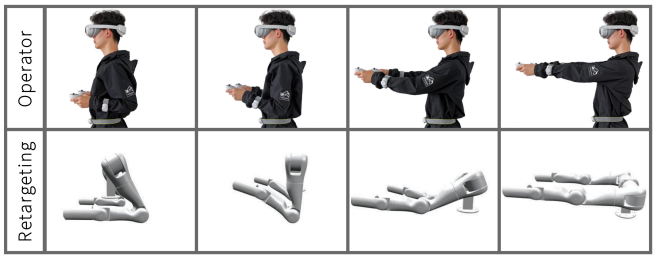


(a) Representative dynamic anthropomorphic poses

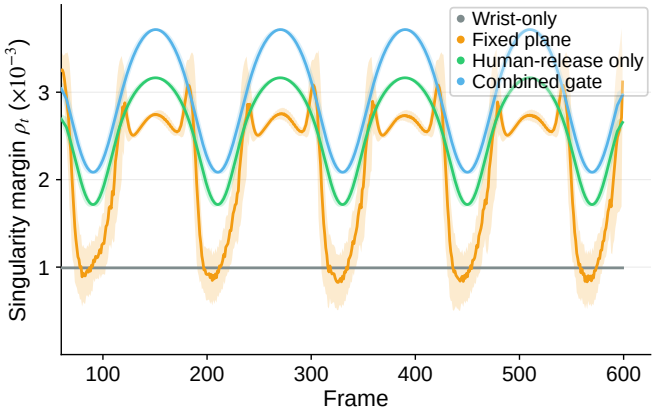


(b) Swivel-angle error trajectories

Fig. 5. Dynamic anthropomorphic behavior. (a) Human and robot arm configurations at selected frames of the elbow-motion sequence. (b) Swivel-angle error trajectories of GMR, Fourier-Teleop, XR-Teleop, and Ours.



(a) Forward-extension motion sequence



(b) Frame-wise singularity margin

Fig. 7. Dynamic singularity behavior. (a) Forward-extension motion sequence and robot configurations near full extension. (b) Frame-wise normalized singularity margin ρ_t after the initial transient.

D. Cross-Platform and Real-Robot Validation

Cross-platform validation. We deploy the solver on three distinct humanoid platforms without parameter retuning: Fourier GR-1 (wheeled dual-arm), JAKA K1 (humanoid dual-arm), and Unitree H1 (full-size humanoid). Fig. 6 provides qualitative cross-platform demonstrations across the different morphologies.

Real-robot task deployment. We further deploy the solver on one motion sequence and two bimanual tasks requiring sustained contact, tight inter-arm coordination, and operation near singularities. Fig. 8 shows representative snapshots. The motion sequence covers neutral, extension, near-singular, recovery, and return phases. In *Stack Rings*, both arms coordinate grasping, transport, and placement; in *Peel Cucumber*, one arm holds the object while the other performs a continuous tool stroke. Real-robot demonstrations illustrate the feasibility of the framework for bimanual manipulation.

E. Ablation Experiments and Analysis

Forward-extension stress test. The forward-extension trajectory drives the arm through near-full-extension singular postures. We compare wrist-only retargeting, a fixed swivel target, AEAC alone, and AEAC with singularity-robust mapping. All configurations share the same input, robot model, and control frequency. Table I reports arithmetic means over post-transient frames 60–599. Its singularity column uses the same normalized margin ρ_t and window as Fig. 7(b), not $\min(\sigma)$ or cross-arm sums.

TABLE I

FORWARD-EXTENSION ABLATION.

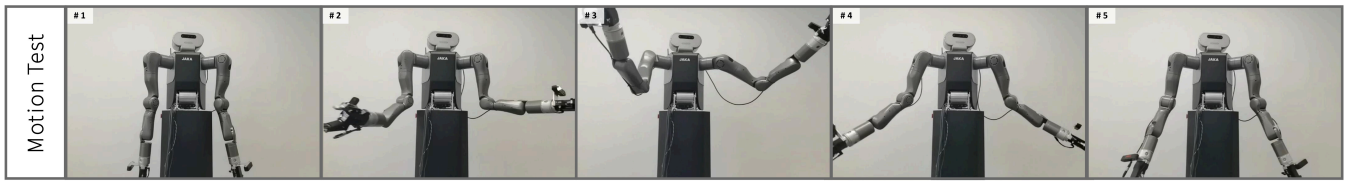
Config.	Post./deg↓	Pos./mm↓	$\rho_t/(10^{-3})$↑	$\bar{\ddot{q}}$/(rad/s^2)↓
Wrist-only	65.65	5.67	0.991	**2.82**
Fixed swivel target	**13.99**	10.27	2.115	168.98
Human-side adaptive	16.88	**5.66**	2.548	3.02
Combined method	17.23	**5.66**	**2.944**	3.20

Post. is anthropomorphic posture error, Pos. is wrist position error, ρ_t is normalized singularity margin, and $\bar{\ddot{q}}$ is mean joint acceleration; "anthro." denotes anthropomorphic.

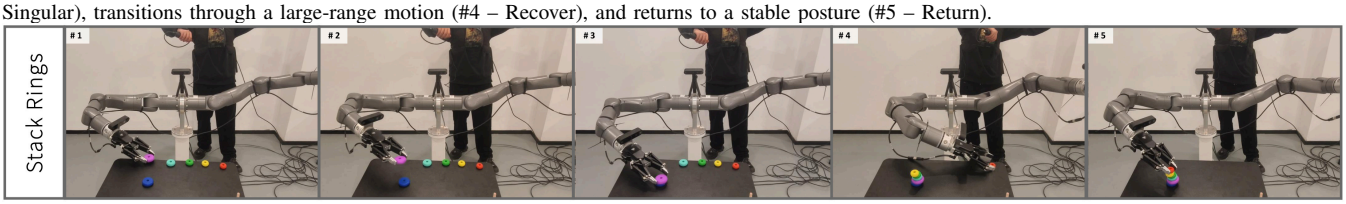
The fixed swivel target reaches a large mean acceleration, 168.98 rad/s^2, because it forces a constant swivel target when elbow evidence is weak. AEAC reduces this mean to 3.02 rad/s^2 and raises mean ρ_t from 2.115×10^{-3} to 2.548×10^{-3}. Adding singularity-robust mapping further raises mean ρ_t to 2.944×10^{-3} while keeping mean acceleration at 3.20 rad/s^2.

Anthropomorphic redundancy resolution. We isolate AEAC by fixing the bimanual wrist targets and changing the desired elbow direction. Figure 9 and Table II show that Ours reduces the mean elbow-direction error from 45.0° to 20.5° while keeping the mean wrist position error at 9.8 mm. The higher-gain setting reduces elbow error further, but increases wrist error and raw joint velocity. Thus, the proposed setting uses the redundant degree of freedom effectively without replacing the primary wrist task.

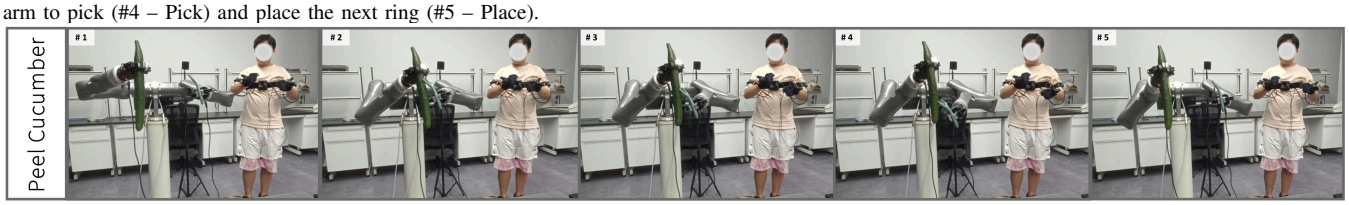
Singularity-robust velocity mapping. We next compare PINV, low- and high-damping least squares, and the pro-



(a) The robot starts from a neutral posture (#1 – Neutral), performs wide arm extension (#2 – Extend), reaches a near-singular raised-arm posture (#3 – Singular), transitions through a large-range motion (#4 – Recover), and returns to a stable posture (#5 – Return).



(b) The robot grasps the first ring (#1 – Grasp), transports it to the target (#2 – Transport), stacks it onto the base (#3 – Stack), and coordinates the other arm to pick (#4 – Pick) and place the next ring (#5 – Place).



(c) The robot secures the cucumber and aligns the peeler (#1, #2 – Align), performs the peeling stroke along its surface (#3, #4 – Peel), and retracts the tool to complete the action (#5 – Retract).

Fig. 8. Real-robot validation. (a) Motion sequence covering neutral, extension, near-singular, recovery, and return phases. (b) Stack Rings: bimanual grasp, transport, and placement. (c) Peel Cucumber: one arm holds the object while the other performs a continuous tool stroke.

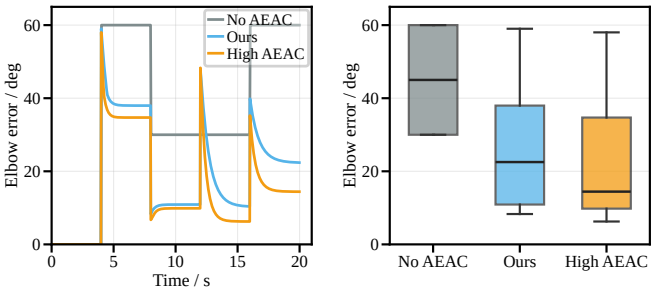


Fig. 9. Anthropomorphic ablation. The proposed AEAC setting reduces elbow-direction error while avoiding the larger motion demand of high gain.

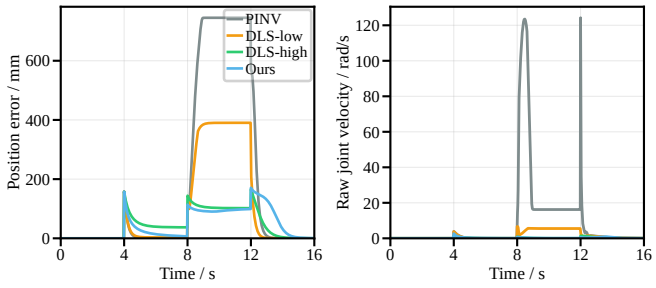


Fig. 10. Singularity ablation. The proposed velocity mapping suppresses raw joint-velocity amplification while maintaining bounded position error near the workspace boundary.

TABLE II

ANTHROPOMORPHIC ABLATION WITH FIXED WRIST TARGETS.

Method	Mean elbow↓	P90 elbow↓	Position↓	$\dot{q}$↓
Ours w/o AEAC	45.0	60.0	0.0	0.00
Ours (default)	**20.5**	38.0	9.8	0.69
Ours (high gain)	16.3	**34.7**	24.3	1.38

TABLE III

SINGULARITY ABLATION ON REACHABLE/UNREACHABLE TRACKING AND VELOCITY DEMAND.

Method	Reachable↓	Unreachable↓	Min. cond.↑	Max. $\dot{q}$↓
PINV	0.0	745.2	0.0030	124.23
DLS-low	1.5	390.3	0.0030	6.66
DLS-high	18.7	101.9	0.0267	1.52
Ours	**4.5**	**97.8**	0.0030	**2.63**

posed J-PARSE mapping on a home–boundary–unreachable–return sequence. Figure 10 and Table III show that PINV produces severe velocity amplification near the unreachable target, whereas Ours limits the peak request to 2.63 rad/s with a 97.8 mm residual. In this test, the mapping balances reachable-target accuracy and peak raw joint-velocity demand.

V. CONCLUSION

This work presents a human-like redundancy-resolution framework for real-time bimanual humanoid teleoperation. SWGR maps XR upper-body motion to robot-scale wrist targets, AEAC converts observable human elbow geometry into an adaptive null-space preference, and a singularity-robust mapping stabilizes the primary wrist task. Experiments on a JAKA K1 across static poses, dynamic trajectories, and bimanual tasks show that the framework improves elbow consistency on redundancy-sensitive motions and reduces abrupt joint motion near singular configurations. One limitation is

that each arm is solved independently, which may limit coordinated bimanual accuracy. Future work will investigate predictive singularity avoidance, inter-arm coordination, and whole-body extensions.

REFERENCES

[1] Z. Luo et al., "SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control," *Science Robotics*, vol. 11, no. 117, Art. no. ead4592, 2026, doi: 10.1126/scirobotics.aed4592.

[2] Y. Ye et al., "Data Pyramid for Embodied Manipulation: A Survey," arXiv:2607.24744, 2026.

[3] X. Cheng et al., "Open-TeleVision: Teleoperation with Immersive Active Visual Feedback," in *Proc. 8th Conf. Robot Learning (CoRL)*, vol. 270, pp. 2729–2749, 2025.

[4] R. Ding et al., "Bunny-VisionPro: Real-Time Bimanual Dexterous Teleoperation for Imitation Learning," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 12248–12255, 2025.

[5] A. J. Elias and J. T. Wen, "Redundancy Parameterization and Inverse Kinematics of 7-DOF Revolute Manipulators," *Mechanism and Machine Theory*, vol. 204, Art. no. 105824, 2024.

[6] Y. Zhao, Y. Tang, J. Zhang, and H. Yu, "Empowering Robot Teleoperation: Exploring the Synergies Between Devices and Manipulator Controllers in a Comparative Study," arXiv:2511.07720, 2025.

[7] C. Kong et al., "A Closed-Form Geometric Retargeting Solver for Upper Body Humanoid Robot Teleoperation," arXiv:2602.01632, 2026.

[8] X. Lin et al., "Human-as-Humanoid: Enabling Zero-Shot Humanoid Learning from Ego-Exo Human Videos with Human-Aligned Embodiments," arXiv:2606.32009, 2026.

[9] M. Shimizu, H. Kakuya, W.-K. Yoon, K. Kitagaki, and K. Kosuge, "Analytical Inverse Kinematic Computation for 7-DOF Redundant Manipulators With Joint Limits and Its Application to Redundancy Resolution," *IEEE Trans. Robot.*, vol. 24, no. 5, pp. 1131–1142, 2008.

[10] S. Li et al., "A Real-Time Explicit Mapping and Teleoperation Control Method for Humanoid Robots with Posture Constraints," *Mechatronics*, vol. 90, Art. no. 102937, 2023.

[11] Z. Yang et al., "An Optimization-Based Scheme for Real-Time Transfer of Human Arm Motion to Robot Arm," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 12220–12225, 2024, doi: 10.1109/IROS58592.2024.10802348.

[12] J. Xiang et al., "A Monocular Vision-Based Robotic Arm Teleoperation Method for Human Arm Configuration Imitation," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 15517–15524, 2025.

[13] B. Chen et al., "HL-IK: A Lightweight Implementation of Human-Like Inverse Kinematics in Humanoid Arms," arXiv:2509.20263, 2025.

[14] J. Zhou et al., "Singularity-Avoidant Motion Mappings for Pose-Tracking Teleoperation With Operational Jacobian Analysis," *IEEE/ASME Trans. Mechatronics*, early access, 2026.

[15] J. P. Araujo et al., "Retargeting Matters: General Motion Retargeting for Humanoid Motion Tracking," arXiv:2510.02252, 2025.

[16] Unitree Robotics, "xr_teleoperate," GitHub repository, [Online]. Available: `https://github.com/unitreerobotics/xr_teleoperate`. Accessed: Sep. 9, 2026.

[17] FFTAI, "Teleoperation," GitHub repository, [Online]. Available: `https://github.com/FFTAI/teleoperation`. Accessed: Sep. 9, 2026.

[18] Z. Zhao, L. Yu, K. Jing, and N. Yang, "XRoboToolkit: A Cross-Platform Framework for Robot Teleoperation," arXiv:2508.00097, 2025.

[19] J. Li, L. Yang, and A. D. Ames, "MIRROR: Visual Motion Imitation via Real-Time Retargeting and Teleoperation with Parallel Differential Inverse Kinematics," arXiv:2603.23995, 2026.

[20] Z. Chen et al., "WARP: Whole-Body Retargeting for Learning from Offline Human Demonstrations," arXiv:2606.29940, 2026.

[21] Y. Shi et al., "A Hybrid Inverse Kinematics Framework for Biomimetic Redundancy Resolution in 7-DoF Humanoid Arms," *Biomimetics*, vol. 11, no. 6, Art. no. 408, 2026.

[22] Y. Lai, Q. Gao, Z. Liang, J. Hu, and Z. Ju, "ObjRetarget: An Object-Aware Motion Retargeting Framework with Anthropomorphic Arm Constraints and Polyhedral Hand Modeling," arXiv:2607.03828, 2026.

[23] W. Zhu, G. Zhang, L. Qiao, L. Ge, and Y. Li, "GHO-WBC: A Gradient-Based Hierarchical Kinematic Optimization Approach to Enhance the Reachability of a Humanoid Robot," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, pp. 18007–18013, 2025.

[24] M. Grobbel, T. Schneider, D. Flögel, and S. Hohmann, "Real-Time and Accurate Collision-Free Teleoperation via Differentiable Constraint-Based Trajectory Planning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2026, arXiv:2606.08725.

[25] J. Carpentier et al., "The Pinocchio C++ Library: A Fast and Flexible Implementation of Rigid Body Dynamics Algorithms and Their Analytical Derivatives," in *Proc. IEEE/SICE Int. Symp. System Integration (SII)*, pp. 614–619, 2019.

[26] L. Yang et al., "OmniRetarget: Interaction-Preserving Data Generation for Humanoid Whole-Body Loco-Manipulation and Scene Interaction," arXiv:2509.26633, 2025.

[27] Y. Ze, Z. Chen, J. P. Araujo, Z.-A. Cao, X. B. Peng, J. Wu, and C. K. Liu, "TWIST: Teleoperated Whole-Body Imitation System," arXiv:2505.02833, 2025.

[28] T. He, Z. Luo, W. Xiao, C. Zhang, K. Kitani, C. Liu, and G. Shi, "Learning Human-to-Humanoid Real-Time Whole-Body Teleoperation," arXiv:2403.04436, 2024.

[29] T. He et al., "HOVER: Versatile Neural Whole-Body Controller for Humanoid Robots," arXiv:2410.21229, 2024.

[30] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Proc. Robot.: Sci. Syst. (RSS)*, 2023.

[31] Y. Ze et al., "TWIST2: Scalable, Portable, and Holistic Humanoid Data Collection System," arXiv:2511.02832, 2025.

[32] T. Yoshikawa, "Manipulability of Robotic Mechanisms," *Int. J. Robot. Res.*, vol. 4, no. 2, pp. 3–9, 1985.

[33] Y. Nakamura, *Advanced Robotics: Redundancy and Optimization*. Addison-Wesley, 1991.

[34] O. Khatib, "A Unified Approach for Motion and Force Control of Robot Manipulators: The Operational Space Formulation," *IEEE J. Robot. Autom.*, vol. 3, no. 1, pp. 43–53, 1987.

[35] S. Chiaverini, "Singularity-Robust Task-Priority Redundancy Resolution for Real-Time Kinematic Control of Robot Manipulators," *IEEE Trans. Robot. Autom.*, vol. 13, no. 3, pp. 398–410, 1997.

[36] A. A. Maciejewski and C. A. Klein, "Obstacle Avoidance for Kinematically Redundant Manipulators in Dynamically Varying Environments," *Int. J. Robot. Res.*, vol. 4, no. 3, pp. 109–117, 1985.

[37] F. Flacco and A. De Luca, "Control of Redundant Robots Under Hard Joint Constraints: Saturation in the Null Space," *IEEE Trans. Robot.*, vol. 31, no. 3, pp. 637–654, 2015.

[38] A. Escande, N. Mansard, and P.-B. Wieber, "Hierarchical Quadratic Programming: Fast Online Humanoid-Robot Motion Generation," *Int. J. Robot. Res.*, vol. 33, no. 7, pp. 1006–1028, 2014.

[39] L. Sentis and O. Khatib, "A Whole-Body Control Framework for Humanoids Operating in Human Environments," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, pp. 2641–2648, 2006.

[40] A. Albu-Schäffer et al., "The DLR Lightweight Robot: Design and Control Concepts for Robots in Human Environments," *Ind. Robot*, vol. 34, no. 5, pp. 376–385, 2007.

[41] A. Allshire et al., "VideoMimic: Visual Imitation Enables Contextual Humanoid Control," arXiv:2505.03729, 2025.

[42] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, *Robotics: Modelling, Planning and Control*. Springer, 2009.

[43] S. Guptasarma, M. Strong, H. Zhen, and M. Kennedy III, "J-PARSE: Jacobian-Based Projection Algorithm for Resolving Singularities Effectively in Inverse Kinematic Control of Serial Manipulators," arXiv:2505.00306v5 [cs.RO], 2026.

[44] PICO Immersive Pte. Ltd., "PICO Motion Tracker," [Online]. Available: `https://www.picoxr.com/global/products/pico-motion-tracker`. Accessed: Sep. 17, 2026.

[45] HTC Corporation, "Vive Tracker (3.0)," [Online]. Available: `https://www.vive.com/us/accessory/tracker3/`. Accessed: Sep. 17, 2026.

[46] G. Casiez, N. Roussel, and D. Vogel, "1€ Filter: A Simple Speed-based Low-pass Filter for Noisy Input in Interactive Systems," in *Proc. ACM CHI Conf. Human Factors Comput. Syst.*, pp. 2527–2530, 2012.

[47] K. Shoemake, "Animating Rotation with Quaternion Curves," in *Proc. ACM SIGGRAPH*, vol. 19, no. 3, pp. 245–254, 1985.